Software Engineering Lab 1

Requirements Engineering & UML Use-Case Modelling

Problem Statement #45 — Multi-Cloud Asset Cost Optimization Portal

SRN: PES1UG24AM282

Domain: Developer Tools & IT Operations

Problem Overview: A FinOps analytics portal that ingests AWS, Azure, and GCP billing CSVs, identifies unattached storage volumes and idle compute instances, and dispatches budget breach alerts.

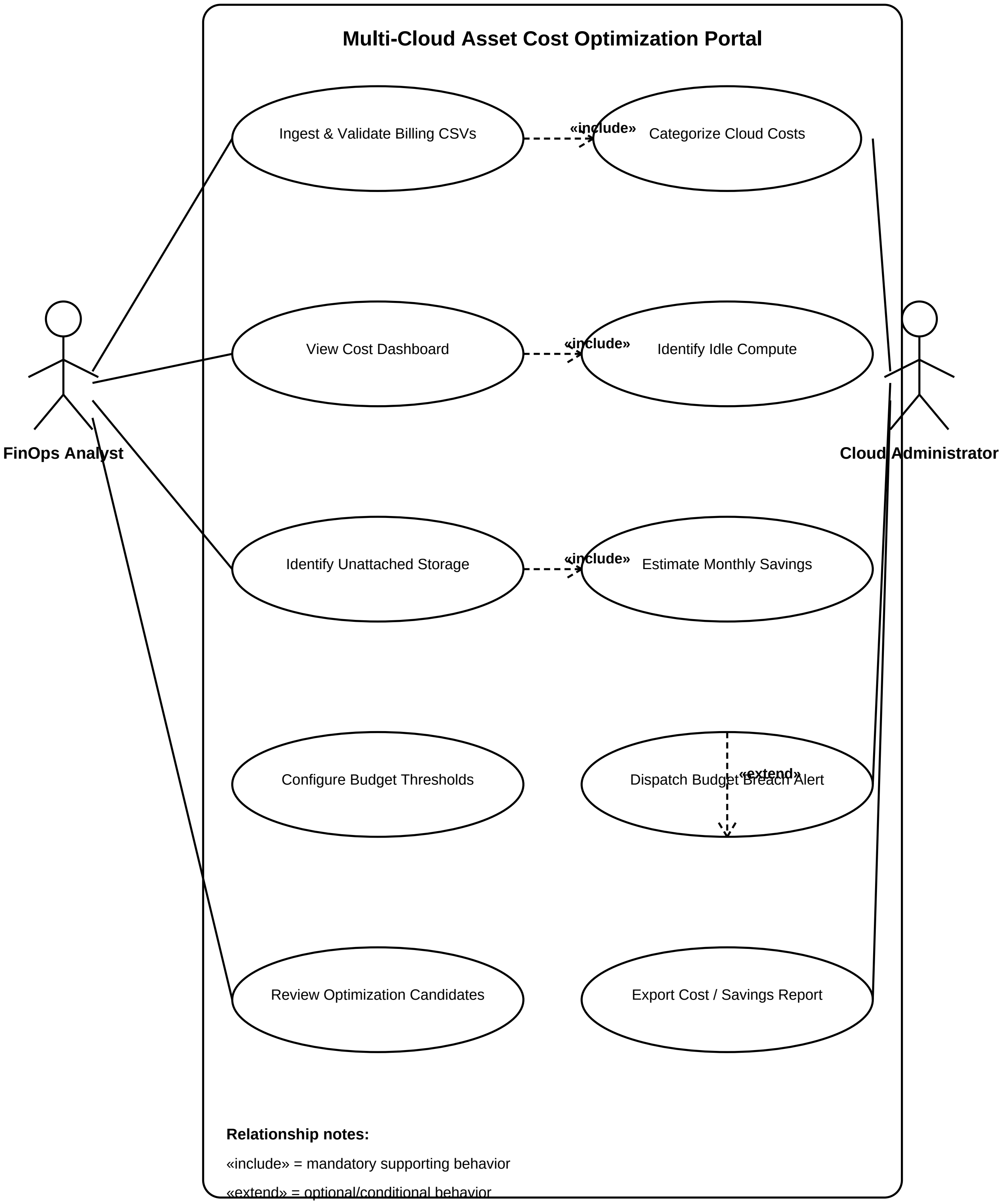
1. Complete Requirements Table

ID	Type	Priority	Description	Acceptance Criteria	Rationale
FR-001	Functional	High	The system shall ingest AWS, Azure, and GCP billing CSV exports and validate required fields before importing the data.	Pass: Valid files from all three providers are imported. Fail: Invalid mandatory fields are silently accepted.	Provides the base data for cost analysis.
FR-002	Functional	High	The system shall categorize imported cloud costs by provider, service, account/project tag, and billing period.	Pass: Dashboard totals can be filtered by these categories. Fail: A cost record is assigned incorrectly.	Enables analysis of spending sources.
FR-003	Functional	High	The system shall identify idle compute instances using <5% CPU utilization over 14 days and estimate potential monthly savings.	Pass: Qualifying instances are flagged with savings estimates. Fail: Active instances are incorrectly classified.	Turns utilization data into optimization opportunities.
FR-004	Functional	High	The system shall identify unattached cloud storage volumes and present them with provider, resource identifier, age, and estimated monthly cost.	Pass: Unattached volumes appear with details and cost. Fail: Attached volumes are incorrectly flagged.	Targets avoidable storage expenditure.
FR-005	Functional	High	The system shall allow administrators to configure budget thresholds and dispatch alerts when monitored spending breaches a configured threshold.	Pass: A breach generates a relevant alert. Fail: No alert is generated after a confirmed breach.	Provides proactive budget control.
NFR-001	Performance & Scalability	High	The portal shall parse multi-gigabyte cloud billing files and update relevant cost dashboards in under 2 minutes under the defined normal operating load.	Pass: Benchmark tests meet the 2-minute target. Fail: Target latency is exceeded.	Supports practical FinOps analysis at enterprise data volumes.
NFR-002	Security	High	The portal shall protect uploaded billing and configuration data through authenticated access, role-based authorization, and encryption in transit and at rest.	Pass: Unauthorized users cannot access protected data/settings. Fail: Protected data is accessible without authorization.	Cloud billing data may contain sensitive financial and operational information.

Requirement count check: Exactly 5 Functional Requirements and 2 Non-Functional Requirements.

2. UML Use-Case Diagram

The diagram below gives the UML page substantially more coverage while keeping the model readable. It includes both required actors, the primary portal functions, three «include» relationships, and one «extend» relationship.



3. Use-Case Flow Specification

Core Use Case: Identify Idle Compute Instances & Estimate Savings

Primary Actor: FinOps Analyst

Goal: Identify compute instances with less than 5% CPU utilization over a 14-day observation period and provide an estimated monthly savings opportunity.

Preconditions

1. The FinOps Analyst is authenticated and authorized.
2. Billing data has been successfully ingested.
3. CPU-utilization data covering the required 14-day period is available.

Postconditions

1. Eligible instances are marked as idle candidates.
2. Each candidate contains resource details and estimated monthly savings.
3. Optimization candidates are available for review.

Main Success Scenario

1. Analyst opens the compute optimization view.
2. System retrieves compute resources and 14-day CPU-utilization data.
3. System evaluates each instance against the **<5%** criterion.
4. System identifies qualifying instances as idle candidates.
5. System estimates monthly savings using recurring cost data.
6. System displays instances, utilization evidence, and savings estimates.
7. Analyst reviews the candidates and takes an optimization decision.

Alternate Flow — Missing or Incomplete Utilization Data

If an instance does not have a complete 14-day utilization record, the system does not classify it as idle. It marks the record as **Insufficient Data**, excludes it from the idle-savings calculation, and displays the reason to the analyst.

Traceability: This use case operationalizes FR-003 and the problem statement's requirement to flag compute instances below 5% CPU utilization over 14 days with a monthly savings estimate.